MASAKA DIOCESAN EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

JOINT MOCK EXAMINATIONS 2026

SCORING GUIDE

P525/1 CHEMISTRY

ITEM ONE

(a): Data Analysis, Interpretation and Conclusions.

- Reactor 1: $-92 = E_{af} - 730$ ✓L

  $E_{af} = +638 kJmol^{-1}$ ✓Da

- Reactor 2: $-92 = E_{af} - 272$

  $E_{af} = +180\ kJmol^{-1}$ ✓Da

Conclusion: Reactor 2 is the one catalysed by the iron catalyst because its forward activation energy is lowered from $+638 kJmol^{-1}$ to $+180\ kJmol^{-1}$. ✓Di

Determination of the Rate Equation and Reactant Optimization

The general rate law equation is: $Rate = k[N_2]^x[H_2]^y$ ✓L Where x and y are orders with respect to $[N_2]$ and $[H_2]$.

Order with respect to N_2 : Compare experiments 1 and 2 where $[H_2]$ is kept constant at $0.20 moldm^{-3}$:

$$\frac{Rate2}{Rate\ 1} = \frac{8.0 \times 10^{-3}}{4.0 \times 10^{-3}} = \left(\frac{0.4}{0.2}\right)^x$$

$$2 = 2^x;\ x = 1$$ ✓Da

Order with respect to H_2: Compare experiment 1 and 3 where $[H_2]$ is kept constant at $0.20 moldm^{-3}$:

$$\frac{Rate3}{Rate\ 1} = \frac{1.6 \times 10^{-2}}{4.0 \times 10^{-3}} = \left(\frac{0.4}{0.2}\right)^y$$

$$4 = 2^y;\ y = 2$$ ✓Da

$$Rate = k[N_2]^1[H_2]^2$$ ✓Dc

Conclusion: Because the reaction is second-order with respect to hydrogen, doubling its concentration increases the rate ~~by~~ four times. Doubling the concentration of nitrogen would only double the rate. Hydrogen concentration should therefore be doubled. ✓Di Cr

(b) Operating Conditions & Industrial Recommendations

Effects of Operating Conditions on Reaction Rate

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Factor	Effect on Reaction Rate	Kinetic Explanation
Iron Catalyst	Increases rate significantly	Provides an alternative reaction pathway with lower activation energy; more colliding particles possess the required energy to react successfully.
High Pressure	Increases rate	Compresses gases, increasing the number of reactant particles per unit volume. This increases collision frequency.
High Temperature	Increases rate	Increases kinetic energy of particles, leading to more frequent and higher-energy collisions exceeding.

Real-Life Applicability of the rate law

Cost optimization of raw materials

Gases like nitrogen and hydrogen cost money to source, purify, and compress. The shows that the reaction is **second-order** with respect to hydrogen, but only **first-order** with respect to nitrogen. Instead of buying and pumping equal amounts of both gases, the plant will intentionally feed a higher proportion of hydrogen into the system. Because of the second-order mathematical relationship, more money spent on hydrogen yields a much faster production rate than spent on nitrogen.

Recommendation for Efficient Production

Selected Reactor: Reactor 2 must be deployed for industrial production.

Reasoning: Operating Reactor 1 would require extreme, high-cost temperatures to overcome its massive activation energy. Reactor 2 operates efficiently at lower temperatures due to the iron catalyst, protecting equipment, saving massive energy costs.

ITEM TWO

(a) Katrinah's mixture;

Moles of $HNO_3 = \left(\frac{0.35 \times 65}{1000}\right) = 0.02275$

Moles of $NH_3 = \left(\frac{25 \times 0.91}{1000}\right) = 0.02275$

Since mole ratio is 1:1:1 for NH_3 : HNO_3: NH_4NO_3 ,the pH of the solution is due to hydrolysis of NH_4NO_3

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Moles of $NH_4NO_3 = 0.02275$

$NH_4^+(aq) + H_2O(l) \qquad NH_3(aq) + H_3O^+(aq)$

$$K_h = \frac{[NH_3][H_3O^+]}{[NH_4^+]}$$

But at equilibrium; $[H_3O^+] = [NH_3]$ and $[NH_4^+] = [NH_4NO_3]$

Total volume of solution =65+25=90cm^3

$$[NH_4^+] = [NH_4NO_3] = \left(\frac{1000 \times 0.02275}{90}\right) = 0.25278M$$ ✓Da.

From $K_h = \frac{K_w}{K_b} \rightarrow K_h = \frac{1.0\times10^{-14}}{1.8\times10^{-5}} = 5.5556 \times 10^{-10} moldm^{-3}$ ✓L ✓Da.

Therefore $K_h = \frac{[H_3O^+]^2}{0.25278} = 5.5556 \times 10^{-10}$

$$[H_3O^+] = \sqrt{5.5556 \times 10^{-10} \times 0.25278} = 1.185 \times 10^{-5}M$$

$$pH = -\log_{10}[H_3O^+] = -\log_{10}(1.185 \times 10^{-5}) = 4.93$$ ✓Da.

Katrinah's mixture is acidic hence fails the safety requirement pH range of 6.5-8.5.

This is because the salt formed undergoes hydrolysis to form hydrogen ions. ✓Di

Joram's mixture;

Moles of HNO_3=0.02275

Moles of NH_3=$\left(\frac{26.39\times0.91}{1000}\right) = 0.0240149$

Since mole ratio is 1:1:1 for NH_3 : HNO_3: NH_4NO_3,

Moles of excess(unreacted) $NH_3 = 0.0240149 - 0.02275 = 0.0012649$

Moles of excess NH_3 + Moles of salt formed constitute a basic buffer

Total volume of solution =26.39+65=~~90.39cm^3~~ 91.39 cm^3.

$$[NH_3] = \left(\frac{1000 \times 0.0012649}{90.39}\right) = \text{0.013994M}$$ ✓Da 0.01384M.

$$[NH_4NO_3] = \left(\frac{1000 \times 0.02275}{90.39}\right) = \text{0.251687M}$$ ✓Da 0.24893M.

$$pOH = pK_b + \log_{10}\left(\frac{[NH_4NO_3]}{[NH_3]}\right) = -\log_{10}(1.8 \times 10^{-5}) + \log_{10}\left(\frac{\text{0.251687}}{\text{0.013994}}\right) = 5.9996 \approx 6$$ ✓Da

0.24893 / 0.01384 ✓L

$$pH = 14 - pOH = 14 - 6 = 8.00$$ ✓Da.

The pH of the buffer solution is within the safety requirement pH range of 6.5-8.5 hence Joram's suggestion was right. ✓Di

(b) A decrease in concentration increases the degree of dissociation of an acid ✓fe

increasing the ~~number~~ concentration of hydrogen ions in solution hence reducing pH.

For acids that ionize endothermically, increase in temperature increases the degree ✓fe

of dissociation hence increasing the ~~number~~ concentration of hydrogen ions in solution, reducing pH.

Em2 = 2L+2Fe=05 · 2L+1Fe=04 · L=01

2Ac=02

1Ac=01

7Da+2Di=Em2=06

4-6Da+1Di=04

2-3Da=01

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For acids that ionize exothermically, increase in temperature decreases the degree of dissociation hence decreasing the number of hydrogen ions in solution, increasing pH.

Clinical uses of buffers;

A blood buffer is used during dialysis. The blood buffer consists of carbonic acid and hydrogen carbonate ions. It maintains pH of blood in the range 7.35-7.45. If pH falls below the range, acidosis happens and ~~if it rises~~ higher, alkalosis is caused. Any of the two conditions denatures proteins and disrupts cellular enzymes. If hydrogen ions from metabolism enter the blood stream, they react with hydrogencarbonate ions to form weak carbonic acid.

$$H^+(aq) + HCO_3^-(aq) \qquad H_2CO_3(aq)$$

The carbonic acid then dissociates into water and carbon dioxide which is transported to lungs and exhaled. This prevents a large decrease in pH.

If an alkaline substance enters the blood stream, the hydroxide ions react with carbonic acid to form hydrogencarbonate ions and water. This minimizes an increase in blood pH.

$$OH^-(aq) + H_2CO_3(aq) \longrightarrow HCO_3^-(aq) + H_2O(l)$$

Phosphate buffers are used in eye drops to maintain a nearly constant pH. The blood buffer consists of dihydrogenphosphate ions and hydrogenphosphate ions. If hydrogen ions enter the eye drop solution, they react with hydrogenphosphate ions to minimize a decrease in pH.

$$H^+(aq) + HPO_4^{2-}(aq) \longrightarrow H_2PO_4^-(aq)$$

If hydroxide ions enter the eye drops, they react with dihydrogenphosphate ions to minimize an increase in pH.

$$OH^-(aq) + H_2PO_4^-(aq) \longrightarrow HPO_4^{2-}(aq) + H_2O(l)$$

Industrial uses of buffers;

A citric acid-sodium citrate buffer is added to cosmetic products during their manufacture to maintain pH between 4.7-5.7 to maintain skin health and for preservation. If acidic substances add excess hydrogen ions to the cosmetic, they react with citrate ions to form citric acid to minimize a decrease in pH.

If excess hydroxide ions are added, they react with citric acid to form citrate ions and water to minimize an increase in pH.

N.B A phosphate buffer can also be used for the same role.

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During fermentation, microorganisms metabolise nutrients, producing acids such as lactic acid or alkaline by-products. A phosphate buffer or citrate buffer is used to maintain pH almost constant as already explained above.

ITEM 3

Atomic radius **increases** from beryllium to barium. This is because from beryllium from barium, **nuclear charge increases as protons are being added to the nucleus as you move from one element to the next. Screening effect also increases,** because an **extra inner energy level completely filled with electrons** is added. **Increase in screening effect outweighs increase in nuclear charge. Effective nuclear charge decreases,** outer most electrons far away and weakly attracted. Melting point **generally decreases from beryllium to barium with magnesium having an abnormally lower melting point than calcium.**

This is because from one element to another, **atomic radius increases** down the group, **the delocalized electrons get further from the positively charged nucleus.** Therefore, **the strength of electrostatic forces of attraction between the positive nucleus and the delocalized electrons generally decrease.** The melting point of magnesium is **lower than that of calcium** because the two metals have a **different crystalline structure.**

Recommendations

Magnesium is the best metal used to manufacture disposable medical components that require a metal that melts easily and flows into moulds because it has lowest melting point (650°C) among the metals listed.

Beryllium is the best metal for manufacture of components that must withstand moderately high temperatures without melting because it has the **highest melting point** (1287°C).

Reactions with water

Beryllium is unaffected by water at all conditions **because it forms a protective layer of beryllium oxide which prevents water** from reaching the metal surface. Strongly **heated magnesium is oxidised by steam because** to form **magnesium oxide** as the steam is reduced to hydrogen gas.

$$Mg(s) + H_2O\,(g) \longrightarrow MgO(s) + H_2(g)$$

Magnesium, calcium, strontium and barium are all **oxidized** by **cold water slowly to magnesium hydroxide, calcium hydroxide, strontium hydroxide and barium hydroxide respectively** and the water **reduced to hydrogen gas.**

$$M(s) + 2H_2O(l) \longrightarrow M(OH)_2(aq) + H_2(g)$$

Where M =Mg; Ca; Sr *or* Ba.

Reaction with dilute sulphuric acid.

Beryllium **reduces hydrogen ions** from the acid, on **warming** to form **hydrogen gas** as it is **oxidized to beryllium sulphate.**

$$Be(s) + 2H^+(aq) \longrightarrow Be^{2+}(aq) + H_2(g)$$

Or beryllium being **amphoteric**, reacts with **warm** dilute sulphuric acid to form **beryllium sulphate and hydrogen gas.**

$$Be(s) + H_2SO_4(aq) \longrightarrow BeSO_4(aq) + H_2(g)$$

Magnesium **reduces hydrogen ions** from the **cold** acid to form **hydrogen gas as it is oxidized to magnesium sulphate.**

$$Mg(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2(g)$$

Or magnesium being **basic**, reacts with **cold** dilute sulphuric acid to form **magnesium sulphate and hydrogen gas.**

$$Mg(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2(g)$$

Calcium, strontium and barium are **basic**, the react with **cold** dilute sulphuric acid for a **short time and the reaction stops.** This is because the each of the metals forms a **sparingly soluble** sulphate that forms a coating on the metal. This prevents further contact between the metal and the acid for the reaction to progress. A little amount of hydrogen gas is formed.

$$Y(s) + H_2SO_4(aq) \longrightarrow YSO_4(s) + H_2(g)$$

Y = Ca; Sr *or* Ba

Explained other application(s), explained dangers and explained mitigations of any of beryllium and magnesium (the selected elements)

Beryllium

Application	Explanation
X-ray tube windows	Beryllium has a **low atomic number**, allowing X-rays to pass through with **very little absorption** so it is used as the **windows of X-ray tubes**, enabling most of the X-rays to reach the

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	patient without significant energy loss, resulting in clear, high-quality diagnostic images.
Copper-beryllium electrical contacts and springs	Beryllium forms alloys that combine **high strength**, and **good electrical conductivity**. These alloys are used to make electrical contacts and springs that retain their shape while conducting electricity resulting into reliable and long-lasting electrical and electronic devices. √Ea

Dangers and Mitigations of using Beryllium

Explained danger	Explained mitigation
Chronic beryllium disease: During **machining or welding, fine beryllium dust** is produced which when **inhaled by workers** into the lungs result into **permanent breathing difficulties** and **reduced lung function**. √Ed	Using **HEPA (High-Efficiency Particulate Air filters)** which **prevent beryllium dust** from becoming airborne or **entering the lungs**, thereby breaking the exposure pathway that causes the disease. √Em
Lung cancer: **Prolonged exposure** to **airborne beryllium particles** allows the particles to **accumulate in the lungs** where **they damage DNA** and interfere with normal cell division, resulting in **uncontrolled growth of lung cells and cancer**. √Ed.	

Explained applications of magnesium

Application	Explanation
Vehicle and aircraft components	Magnesium has a **low density** and a **high strength-to-weight ratio** thus is alloyed and used to manufacture vehicle and aircraft parts reducing the overall mass of the structure without greatly reducing strength. √Ea
Sacrificial anodes	Since magnesium is **more reactive than iron and steel**, it is attached to pipelines, ships and water heaters where it oxidises preferentially instead of the protected metal components, resulting in corrosion protection and a longer service life of the structures. √Ea

2Pe = 1L3 = L2 = 04
1Pe = 1L3 = L1 = 02

4Re+4eq = 1C3 = L3 = 08
(2-3)Re+(2-3)eq = L2 = 06
1Re = L1 = 02.

3Ea+2Ed+2Em = 1L3 = L3 = 06
2Ea+2Ed = L2 = 03
2Ea = L3 = 01

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Flares and fireworks	Magnesium reacts vigorously with oxygen and burns with an **intense white flame** which is used in flares and fireworks producing a brilliant light that can be seen over long distances for effective emergency signaling or spectacular fireworks displays. √Ba

Dangers and Mitigations of using magnesium

Explained danger	Explained mitigation
Magnesium fires: Fine magnesium powder or hot magnesium exposure to heat or sparks makes it to react rapidly with air, producing an extremely hot white flame resulting in severe fires, burns and property damage. √Ed	Using **Class D dry powder extinguishers** to eliminate ignition sources by cutting off oxygen supply to stop the combustion process. √Em
Explosion when water is used on burning magnesium: When water is accidentally poured onto burning magnesium, the hot metal reacts with it to produce hydrogen gas which ignites immediately, resulting in a violent explosion that spreads the fire and causes serious injuries. √Ed	Use **Class D extinguishers** instead. Dry powder isolates the burning metal from oxygen without reacting to produce hydrogen gas, preventing the explosion. √Em ·

ITEM 4

The physical properties of the selected elements include:

(a) Trend in electronegativity;

Electronegativity **decreases** from fluorine to iodine. This is because from fluorine to iodine, **nuclear charge increases due addition of protons to the nucleus** from element to element, **screening effect increases because an extra inner energy level completely filled with electrons is added** from element to element. **Increase in screening effect outweighs increase in nuclear charge**, effective nuclear charge decreases, atomic radius increase and the bonding electrons experience a **greater repulsion than nuclear attraction.** √Pe

(b) Melting point.

Melting point **increases** from fluorine to iodine because **atomic radius increases**, molecular size and **molecular mass increases**, hence **the magnitude of van der Waals forces of attraction holding the molecules increases** hence the energy **required to break the forces therefore increase** from fluorine to iodine. √Pe

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The research team also observed that the halogens differ in their reactions with

(a) Magnesium metal

Dry fluorine **oxidises** magnesium even without heating **to magnesium fluoride.** ✓Re

Dry chlorine, bromine, and iodine **oxidize heated** magnesium metal form the corresponding magnesium halides. ✓Re

$$Mg(s) + F_2(g) \rightarrow MgF_2(s) \checkmark Eq$$

$$Mg(s) + X_2(g/l/s) \rightarrow MgX_2(s) \checkmark \text{Eq where } X = Cl, Br \text{ or } I$$

(b) Hydrogen gas

Fluorine **oxidizes** rapidly in sunlight or when sparked at room temperature or even explosively in darkness forming hydrogen fluoride. ✓Re

$$H_2(g) + F_2(g) \rightarrow 2HF(g) \checkmark Eq$$

Chlorine **oxidizes** hydrogen in the **presence of ultraviolet light** (or direct sunlight) to form **hydrogen chloride gas**

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g) \checkmark Eq$$

Bromine **oxidizes** ✓Re hydrogen gas in **presence of heat** and platinum as a catalyst to form **hydrogen bromide gas.**

$$H_2(g) + Br_2(l) \rightarrow 2HBr(g) \checkmark Eq$$

Iodine **oxidizes** hydrogen gas in **presence of strong heating** and **platinum catalyst** to form **hydrogen iodide gas.** It is a reversible reaction. ✓Re

$$H_2(g) + I_2(s) \rightleftharpoons 2HI(g)$$

(c) concentrated potassium hydroxide solution.

Fluorine **oxidizes hot** concentrated potassium hydroxide solution to form **oxygen** and water and itself reduced to **potassium fluoride** ✓Re

$$2F_2(g) + 4KOH(aq) \rightarrow O_2(g) + 4KF(aq) + 2H_2O(l) \checkmark Eq$$

Chlorine, bromine and iodine **disproportionate** in hot concentrated potassium hydroxide solution to form corresponding potassium halides, potassium halite(v) salts **and water.**

$$3Cl_2(g) + 6KOH(aq) \rightarrow 5KCl(aq) + KClO_3(aq) + 3H_2O(l) \checkmark Eq$$

$$3Br_2(l) + 6KOH(aq) \rightarrow 5KBr(aq) + KBrO_3(aq) + 3H_2O(l)$$

$$3I_2(s) + 6KOH(aq) \rightarrow 5KI(aq) + KIO_3(aq) + 3H_2O(l)$$: accept ionic equations.

Applications of chlorine and fluorine.

Element	Application	Explanation

Fluorine	Manufacture of a high-performance polymer used for insulating electrical cables	Fluorine forms **very strong carbon-fluorine (C-F) covalent bonds**. The strong C-F bonds produce polymers that are thermally stable, chemically inert, and excellent electrical insulators. Therefore, fluorinated polymers such as PTFE are suitable for insulating electrical cables because they prevent the flow of electricity and withstand high temperatures and corrosive chemicals.✓Ea ✓Ea
Chlorine	Used in water treatment	Chlorine is a **strong oxidizing agent**. When chlorine dissolves in water, it forms hypochlorous acid (HOCl), which is a powerful oxidizing agent so it destroys harmful microorganisms by **oxidizing** their cell components and enzymes, thereby **disinfecting the water**. Therefore, a small amount of chlorine remains in the water, providing continued protection against microbial contamination during storage and distribution.✓Ea ✓Ea

Dangers and mitigations of chlorine and fluorine.

Element	Danger	Mitigation
Fluorine	✓ Fluorine gas is extremely poisonous. Inhaling it irritates and damages the nose, throat, and lungs, leading to breathing difficulties and severe lung injury.✓Ed ✓Ed	✓ Handle fluorine only in closed systems with good ventilation. Workers should wear appropriate personal protective equipment (PPE) such as respirators, gloves, and goggles to prevent the inhalation of fluorine gas.✓Em ✓Em

1u4 2Pe =

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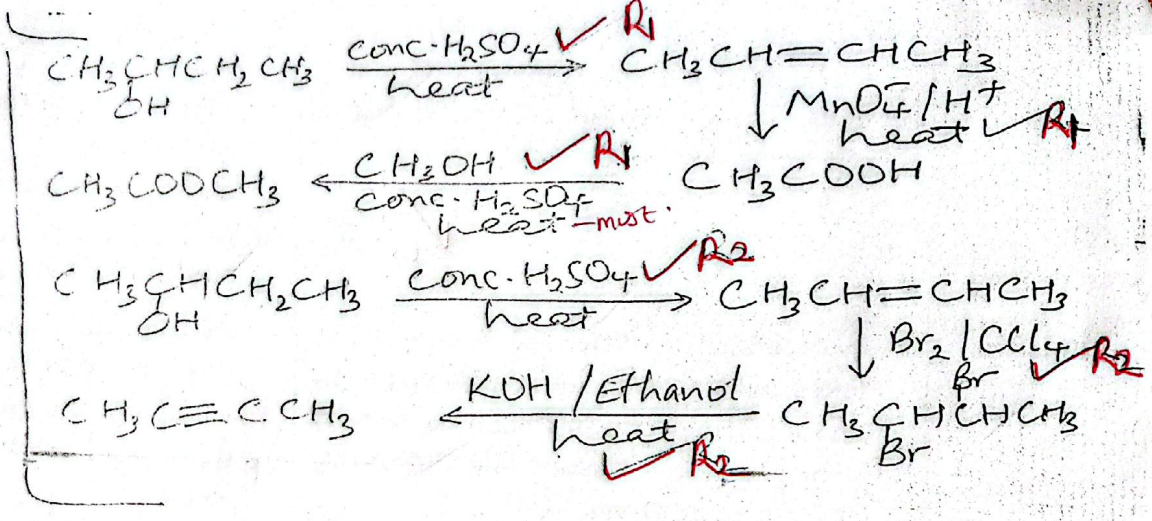
	✓ Fluorine reacts vigorously with many materials, including moist skin and eyes, causing severe chemical burns and permanent eye damage.✓Ed	✓ Use corrosion-resistant equipment, wear chemical-resistant protective clothing, gloves, and face shields to avoid body contacts.✓Em
Chlorine	✓ Excess chlorine released into rivers or lakes kills beneficial microorganisms and aquatic organisms such as fish by damaging their tissues and disrupting aquatic ecosystems.✓Ed	✓ Dechlorinate wastewater before discharge and carefully control the amount of chlorine used. Regularly monitor chlorine concentrations in treated water.✓Em
	✓ Chlorine gas is poisonous. When inhaled, it reacts with moisture in the respiratory tract to form acids which irritate and damage the lungs, causing coughing, chest pain, breathing difficulties, and, in severe cases, pulmonary oedema.✓Ed	✓ Use chlorine in well-ventilated areas or closed systems. Workers should wear appropriate respiratory protective equipment, and chlorine gas detectors should be installed to detect leaks early.✓Em

ITEM FIVE

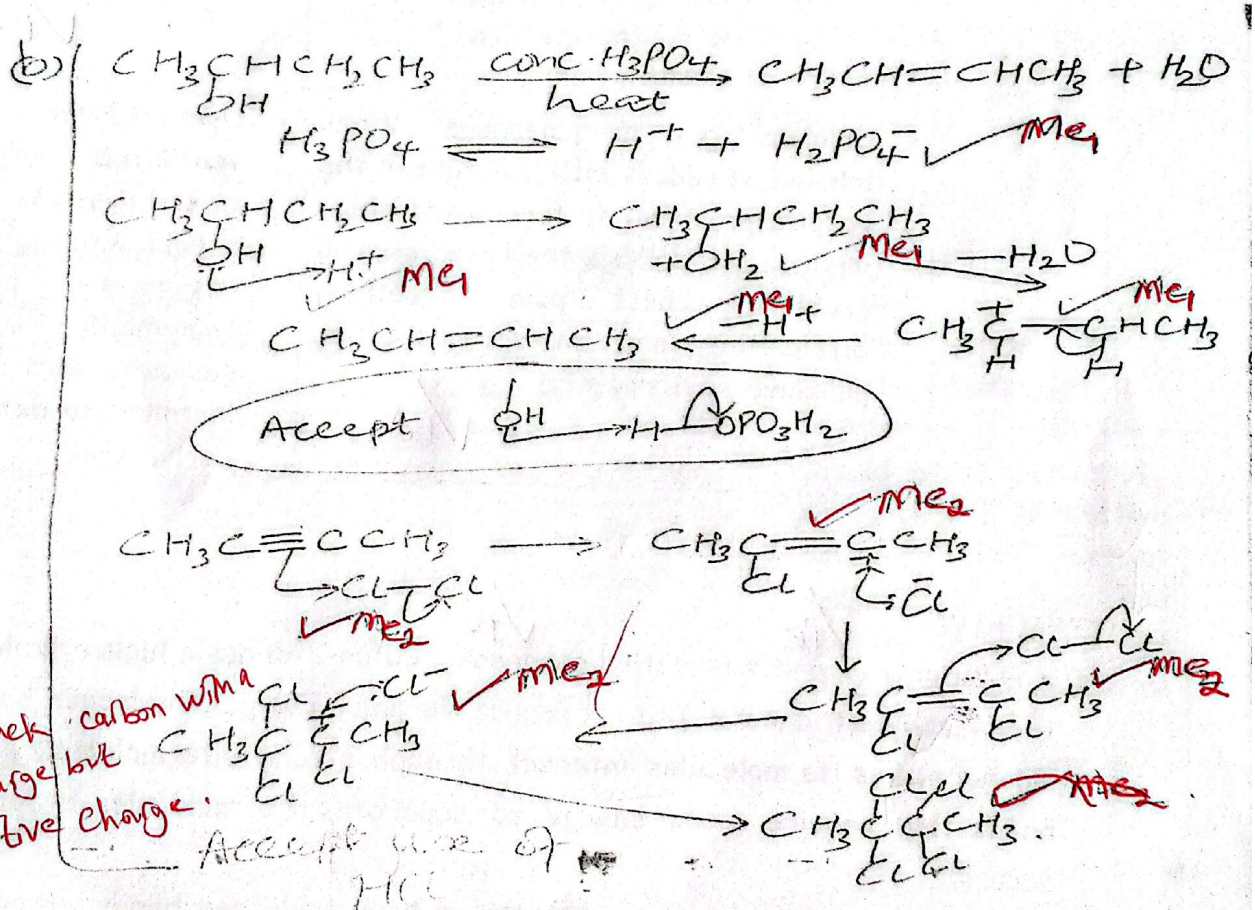
(a) X is butan-2-ol and Y is methyl ethanoate. Butan-2-ol has a **higher boiling point than methyl ethanoate**. This is because the **polar Oxygen-hydrogen bond** in the alcohol makes its **molecules interact through strong intermolecular hydrogen bonds** that **require more energy to separate the molecules** before boiling occurs.

However, methyl ethanoate molecules **cannot form hydrogen bonds** but are held by **weaker** dipole-dipole attractions/London dispersion forces/ **van der waals' forces** that require a **lower amount of energy** to break.

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(b)



UQs 3Pe = L2 = 04
1/2Pe = L1 = 02

OMs ~~Emitmet~~ 5m1 + 4m2 = L2 = 08
3/4m1 / 3m2 = L1 = 03

ORs 3R1 + 3R2 = L2 = 08
2R1 / 2R2 = L1 = 03

OEs 2Ea + 2Ed + 2Ed = L3 = (
2Ea + 2Ed = L2 = 03 (12)
2Ea = L1 = 01

butane

2,2,3,3-Tetrachloromethane is used in making heavy duty engine degreasers because it is highly non-polar hence readily dissolves in non-polar substances such as oils, grease, lubricants, etc.

During its use, its high chlorine content makes it resistant to burning hence not easily flammable/chemically stable for use on hot engines.

However, because it is a very strong non-polar solvent, it can dissolve plastics, rubber seals, paints, protective coatings and can only be used to clean metal components/ destroys non-metallic engine components.

The compound contains carbon-chlorine bonds that resist natural decomposition hence non-biodegradable, remaining in the environment for a long time hence harming plants and animals if improperly disposed off.

Being lipophilic (fat soluble), it can accumulate in the tissues of fish and other aquatic organisms hence becoming concentrated in the food chain risking biodiversity if not disposed off well.

The above dangers can be mitigated by using closed-loop degreasing systems that capture and recycle solvent vapours to prevent the compound from escaping to the environment.

ITEM 6

Suitability of the suggested products.

- Ethylbenzene is used as fuel additive because it has **high octane rating preventing engine knocking** by ensuring smooth, controlled combustion instead of premature, uncontrolled ignition inside internal combustion engines.

Ethylbenzene contains a **stable aromatic benzene ring, which makes it more resistant to premature oxidation** under the high temperature and pressure inside an engine cylinder.

Have strong C-H bonds to prevent premature oxidation and high energy on combustion.

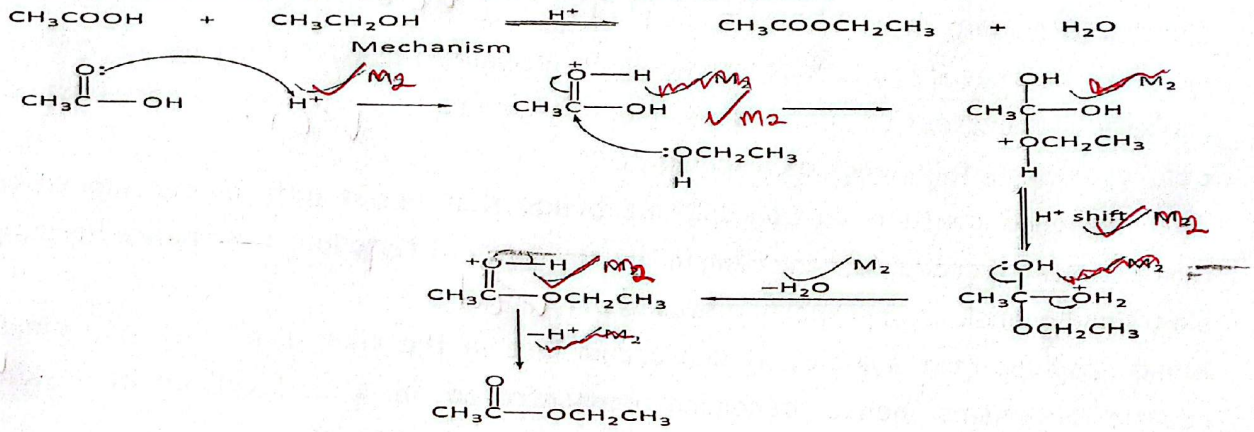
Polypropene is used for making food containers because its **strong/tough/good mechanical strength/high melting point** resists **breaking and deformation during handling/increases the durability and lifespan of the container/can withstand high temperatures of hot food.**

- It has **strong carbon-carbon covalent bonds** and require a **lot of heat to break making it resistant to heat** hence suitable for holding hot foods.

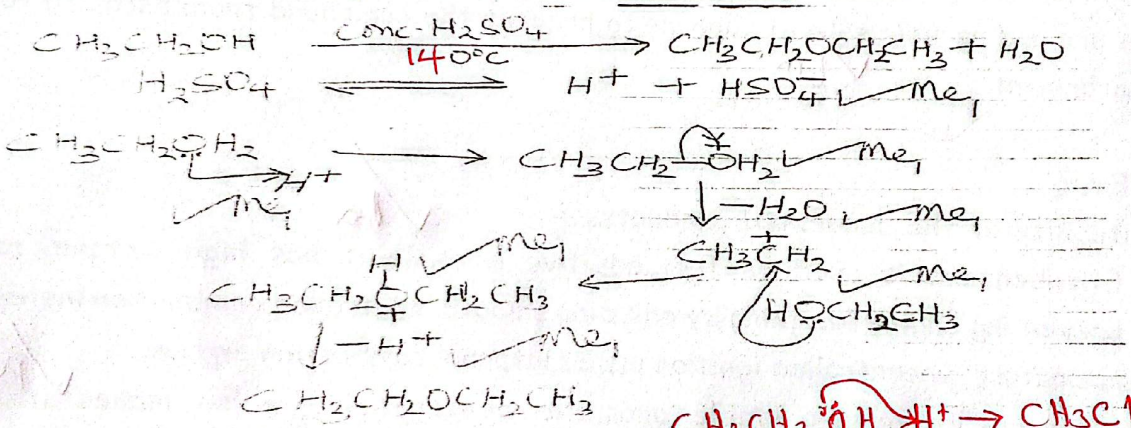
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- Polypropene also consists of **long-chain polymer molecules and has a high molecular mass**, which increase strength of intermolecular forces giving the polymer **a relatively high melting point.**

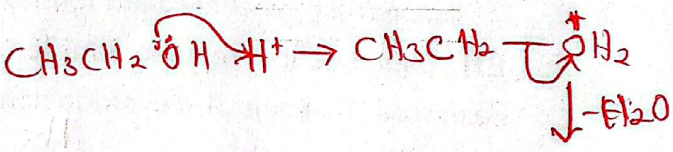
Mechanism for converting left over acid into an ester.

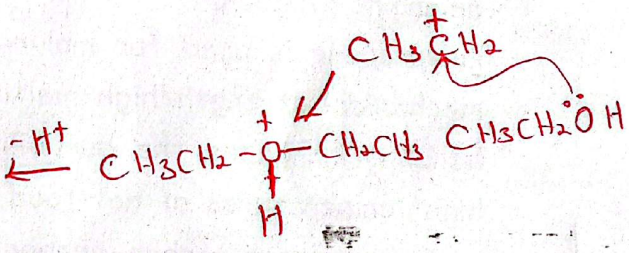


Mechanism for converting leftover ethanol into diethyl ether.



Formation of ethylbenzene from ethanol.



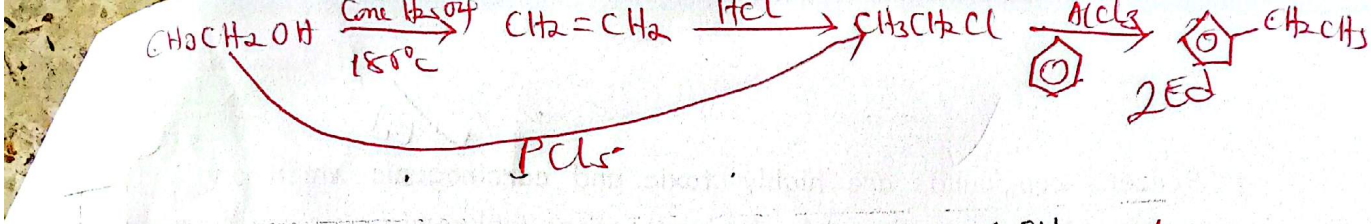


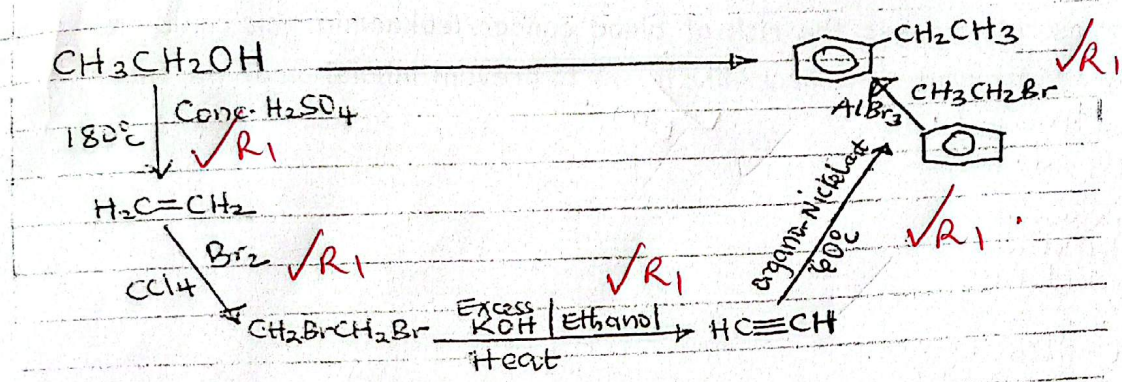
$(CH_3CH_2)_2O$.

3Pe = 04
1/2 = 02.
8m1 + 8m2 = 08
5/6 m1 / 5/6 m2 = 03
3R1 + 5R2 = 08
2R1 / 3R2 = 03.

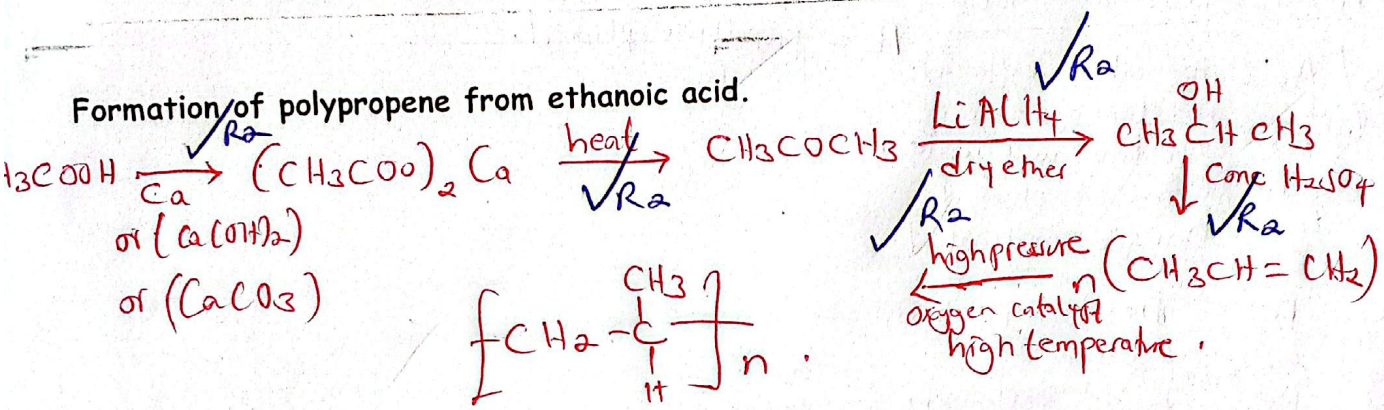
Eaten 2Em = 04
2Ea + 2Ed = 03.
2Ea = 01

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Formation of polypropene from ethanoic acid.



Health implications.

- Prolonged exposure of propene/ethene, which are highly flammable can easily ignite when exposed to flames causing fire and explosions which can lead to death. This can be mitigated by installing gas leak detectors to prevent the leakages.
- Polymerisation reactors operate at high pressure and temperature, posing burn and explosion hazards if poorly controlled. resulting into burns and injuries from hot equipment. This can be mitigated by wearing heat resistant gloves and overalls to prevent the burns.

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✓Ed

- Benzene compounds are highly toxic and carcinogenic which on long-term exposure increases the risk of blood cancer/leukaemia, this can be mitigated by wearing protective gears like masks to prevent inhalation of the fumes.

✓Em.

L - Law / theory / principal.
Fe - Factor explained.
Tm1 - Item 1.
Da - Data analysed.
Di - Data interpleted.
Dc - conclusion of data.
Tei - Item 2.
Ca - Concept applied.
Cr - Concept recommendation.

Pe - Property explained.

③ Ae - Application explained.
④ Re - Reaction explained.
Eq - equation well wrote.
Ed - Danger explained
Em - Danger mitigated.

5/6 Pe - Property Explained.
Eq - Equation.
Re - Reason explained.
Ea - Explained application
Ed - Danger explained.
Em - Danger mitigated.

CH_3 (